

Chapter 1: Energy & Respiration

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Key Outcomes:

- Why Living organisms need energy
- ATP Structure and Why its the universal currency for energy, How ATP is made
- Structure of mitochondria
- The 4 Stages of Aerobic Respiration
- Role of Coenzymes in Aerobic Respiration
- Difference in relative energy values of different respiratory substrates (lipids, carbohydrates, proteins) RQ Value (and calculations)
- Anaerobic Respiration pathways and the differences between them
- Adaptations of rice plants to be able to survive in anaerobic conditions

Definitions

Respiration: The process by which cells release energy from organic molecules, measured as ATP production from ADP and Pi.

ATP (Adenosine Triphosphate): The universal energy currency of cells, consisting of adenine, ribose sugar, and three phosphate groups. Energy is stored in the phosphate bonds.

Anabolic Reactions: reactions that result in building new molecules/compounds (e.g. Protein synthesis, DNA replication, etc...)

Aerobic Respiration: Respiration using oxygen as the final electron acceptor. More efficient than anaerobic respiration, producing ≈ 32 ATP per glucose.

Anaerobic Respiration: Respiration without oxygen. Produces 2 ATP per glucose and allows glycolysis to continue when oxygen is scarce.

Respiratory Quotient (RQ): The ratio of CO₂ produced to O₂ consumed during respiration (RQ = CO₂ out / O₂ in). Varies by substrate

Introduction

Why do living organisms need energy?

For vital processes within the body essential for life, like:

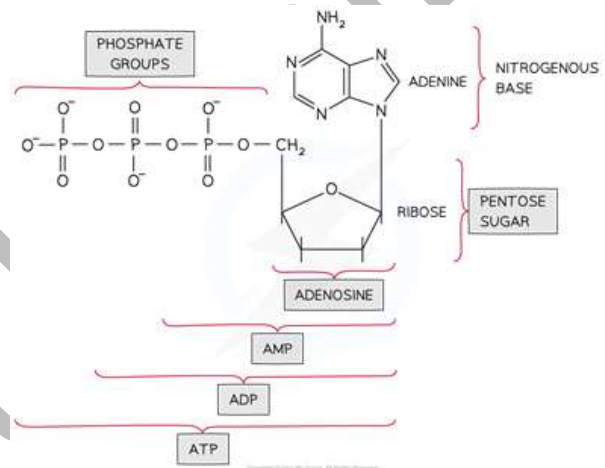
1. **Anabolic Reactions** e.g. Protein synthesis, DNA replication (see definition above)
2. **Active Transport**
3. **Muscle Contractions** for movement
4. **Exocytosis/Endocytosis**
5. **Transmission of nerve impulses**

What is ATP?

Adenosine Triphosphate, Its a molecule that is used as **a universal energy currency**

ATP structure:

- Adenine nitrogen base
- Ribose sugar
- 3 Phosphates (phosphorus + 4 oxygen)



Q: How does the structure of ATP differ from Adenine?

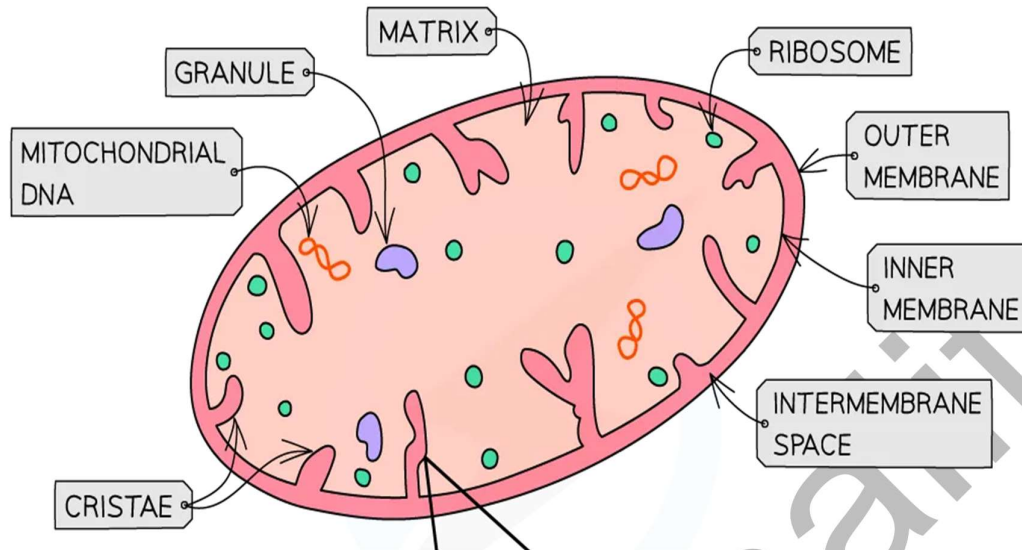
- **ATP has ribose not deoxyribose**
- **ATP has 3 phosphates**

Q: Describe the features that make ATP a suitable universal currency for energy

- Small water soluble molecule
- Hydrolysis leads to **immediate energy release**
- Reversible reaction ($\text{ATP} \rightleftharpoons \text{ADP} + \text{P}_i + 30.5\text{kJ}$)
- Has **High Turnover** (hydrolysis of each phosphate releases **30.5kJ**)
- Link between energy yielding and energy requiring reactions

VERY IMPORTANT QUESTION!!

Mitochondria Structure



Outer membrane:

Permeable to several small molecules (as it contains many transport proteins)

Inner membrane:

- Less permeable than outer membrane (naturally not permeable to protons)
- Site of ETC and ATP synthase (used in oxidative phosphorylation)
- Folded forming Cristae

Intermembrane space:

- Space between the two membranes
- Accumulates Protons (H^+). So, its more acidic than the Matrix

Mitochondrial matrix:

- Like a cytoplasm, but for the mitochondria, contains things like:
 - **Mitochondrial DNA:** Carries genes of necessary proteins and enzymes within a mitochondrion
 - **70S Ribosomes:** For Protein synthesis of respiratory enzymes and proteins within a mitochondrion

***Q:** Explain the importance of the **inner membrane** being folded into cristae

- **Increases the surface area** of **inner membrane**
- SO **Increased number of ETC and ATP synthase**
- So **increases ATP production**

A* Student Question:

Q: Explain why when placed in water, an animal cell bursts while a mitochondrion doesn't burst

- SINCE The mitochondrion **has foldings** in the inner membrane (**cristae**)
- Allowing it to **expand** when water enters/when it becomes turgid
- Animal cell doesn't have a cell wall to protect it from bursting

Aerobic Respiration

How is ATP made?

ATP is made by phosphorylating ADP ($\text{ADP} + \text{P}_i \rightarrow \text{ATP}$)

SO, this means that cells have a constant supply of ADP OR ATP at all times

Throughout the stages of respiration, ATP is made from different reactions and in different amounts.

1. Substrate linked/level phosphorylation:

(reactions where **ATP is a Biproduct** of the reaction), This happens in these stages of respiration

- Glycolysis**
- Krebs Cycle**

2. Oxidative Phosphorylation

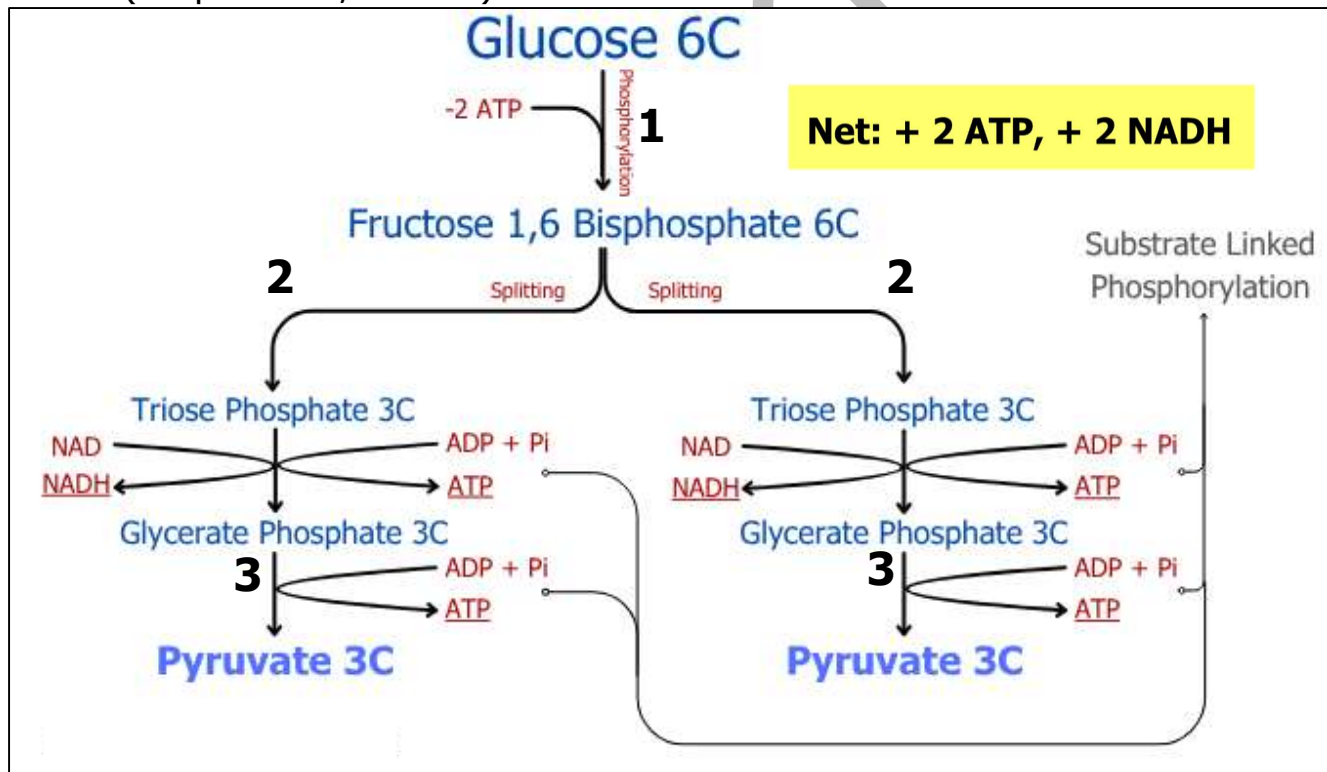
a stage in Aerobic respiration that results in **producing large amounts of ATP** utilizing **ETC, chemiosmosis and ATP Synthase** protein (will be explained further in detail)

Note: Different Substances (carbohydrates, lipids, protein) can be **substrates for respiration**
(REMEMBER THIS FOR LATER)

Stages of Aerobic Respiration (Carbohydrates)

1. Glycolysis (Cytoplasm)

1. **Phosphorylation:** Glucose (6C) is phosphorylated using 2 ATP to form fructose-1,6-bisphosphate (6C), this is to:
 - a. Trap the Glucose molecule in the cell (prevent it from diffusing out)
 - b. Activate the Glucose/Increase its energy levels
2. **Splitting:** Fructose-1,6-bisphosphate splits into TWO 3C Triose phosphates (TP).
3. **Oxidation:** Each Triose phosphate is oxidized to form Glycerate Phosphate (GP), reducing NAD to NADH and producing ATP. Then GP is broken down into Pyruvate (3C) producing ATP.
4. **ATP Production:** Substrate-linked phosphorylation yields a net gain of +2 ATP (+4 produced, -2 used)



Pyruvate molecules then diffuse into the mitochondria where they continue aerobic respiration stages

Q: Explain why glycolysis needs ATP first/ why Glucose gets phosphorylated in the beginning of glycolysis

Slow down! What is this NAD and NADH??

NAD is a coenzyme that is **ESSENTIAL** for respiration, since it:

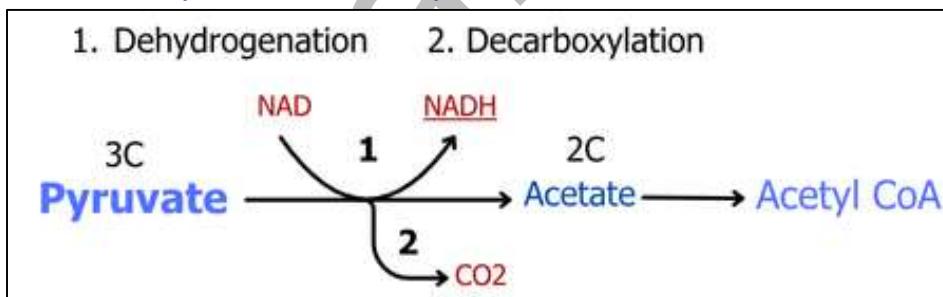
- Carries H atoms from dehydrogenation reactions, gets reduced into **reduced NAD (NADH)**
- Acts as a **Coenzyme to Dehydrogenase** enzyme, responsible for dehydrogenation
- **Transports Hydrogen** Atoms to **Cristae** of mitochondria for ETC (Oxidative phosphorylation)
- Each **NADH** results in production of **2.5 ATP** molecules in **Oxidative Phosphorylation**

FAD Is another Coenzyme just like NAD, but only 1.5 ATP is produced from it in Oxidative phosphorylation

2. Link Reaction (Mitochondrial Matrix)

1. Decarboxylation: **Pyruvate** loses a CO_2 molecule to form an acetyl group, becoming a 2C molecule (by decarboxylase enzyme)
2. Dehydrogenation: **Pyruvate** is oxidized by dehydrogenase enzyme (pyruvate - H), reducing **NAD** to **NADH**, So **Pyruvate** $\rightarrow$ **Acetate**
3. Coenzyme A Binding: The acetyl group combines with coenzyme A to form **Acetyl-CoA**, to join Link Reaction with Kreb's Cycle

*Link Reaction: $2 \text{ Pyruvate} \rightarrow 2 \text{ Acetyl CoA} + 2 \text{ NADH}$



Total So Far:

Glucose $\rightarrow$ 2 Acetyl CoA + 4 NADH + 2 ATP

Exam Tip: The **link reaction** indirectly requires **oxygen** because **NADH** must be reoxidized in the ETC to regenerate **NAD**.

So, No oxygen = No link reaction

(+no **Krebs cycle**, **oxidative phosphorylation**)

Chemistry Reminder:

- Reduction = Gain of hydrogen atom
 $\text{NAD} + \text{H} \rightarrow \text{NADH}$
- Oxidation = loss of hydrogen atom (Regeneration of NAD)
 $\text{NADH} \rightarrow \text{NAD} + \text{H}$

3. Krebs Cycle (Mitochondrial Matrix)

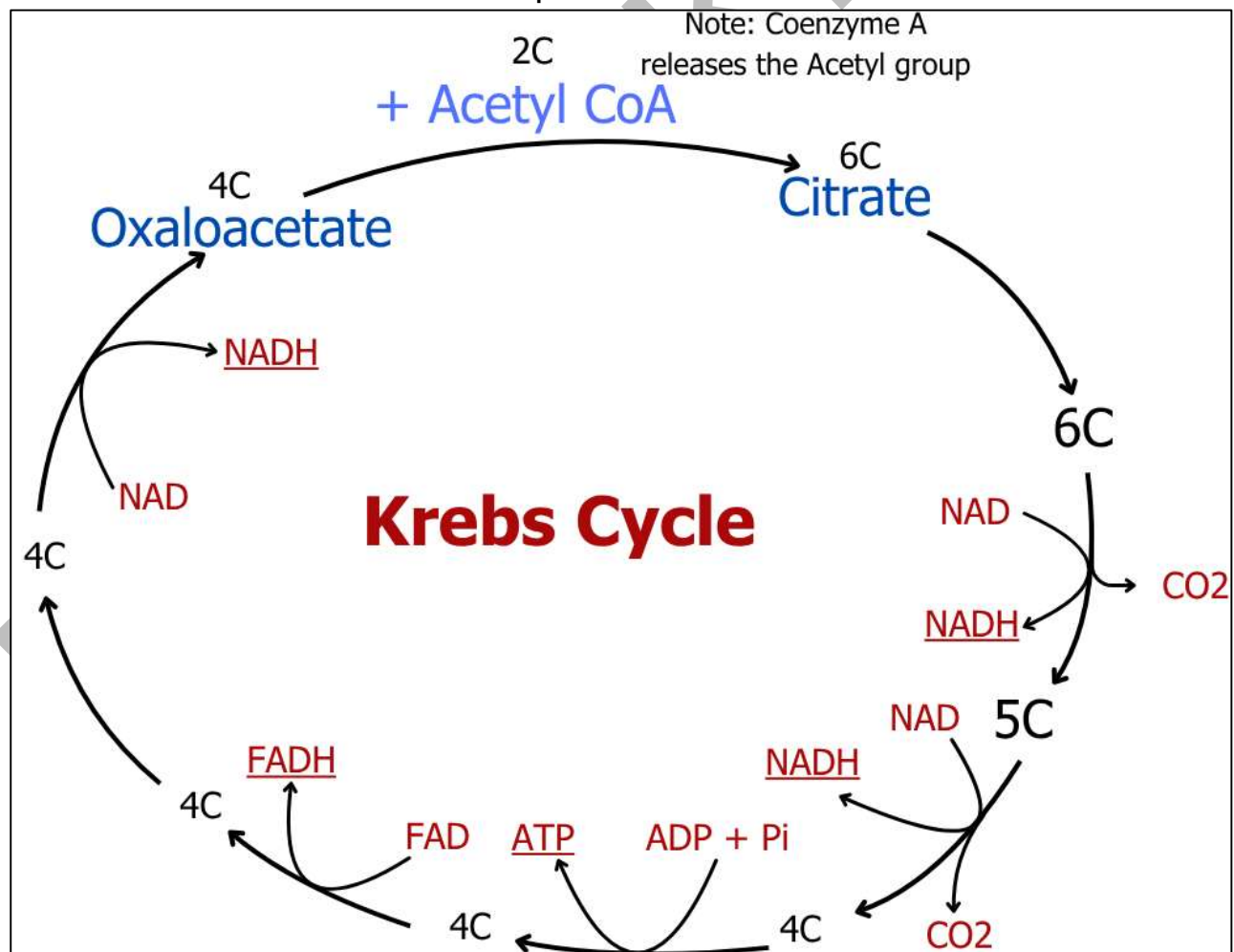
3. **Citrate Formation:** Each **Acetyl-CoA** (2C) combines with **oxaloacetate** (4C) to form **citrate** (6C).
4. **Decarboxylation and Dehydrogenation:** **Citrate** undergoes multiple steps, releasing 2 CO₂ and reducing **NAD/FAD** to **NADH/FADH**.
5. **Substrate-Level Phosphorylation:** **One ATP** is produced per cycle.
6. **Regeneration of Oxaloacetate:** **Oxaloacetate** is **regenerated** to accept another acetyl group.

Remember, 2 Acetyl CoA are produced from link reaction, each enter separate Krebs cycle, so 2x Krebs Cycles occur in respiration of 1 Glucose molecule

Exam Tip: Krebs cycle of 2 Acetyl CoA produces **6 NADH, 2 FADH₂, 4 CO₂ and 2 ATP**. **Oxaloacetate** acts as both starting and ending point.

Q: Explain why **Krebs Cycle** is called a 'cycle'

- A: - It leads to **regeneration of Oxaloacetate**, its the starting and the ending point
- All intermediate molecules are present at all times



Total So Far: 1 Glucose = 10 NADH + 2 FADH + 2 ATP + Oxaloacetate (regenerated) + 4 CO₂ (waste gas)

Fill This Up!

Stage	ATP	CO ₂	NADH	FADH ₂
Glycolysis				
Link Reaction				
Krebs Cycle				
TOTAL				

4. Oxidative Phosphorylation (Inner Mitochondrial Membrane/Cristae of mitochondria)

1. Electron Transport Chain (ETC):

- **(Reduced)NADH/FADH** (from Glycolysis, Link Reaction, Krebs Cycle) Release H atoms at the **inner mitochondrial membrane**, **H splits into Proton and Electron**
- **Electrons** travel **along** protein complexes (ETC), **releasing energy from excited electrons to power proton pump** to pump **protons (H⁺)** into the **intermembrane space**.

2. Chemiosmosis:

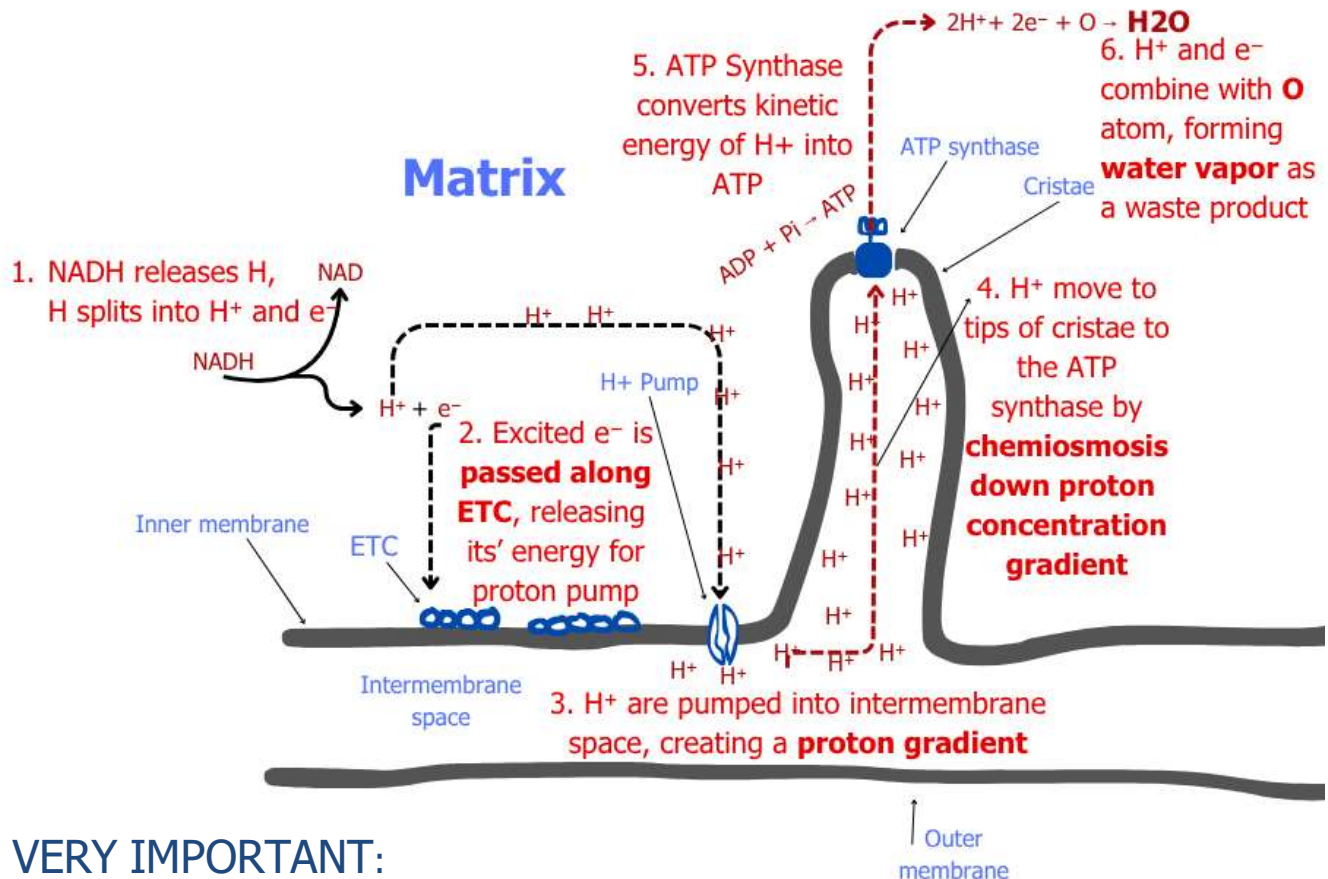
- **Protons** build up in the intermembrane space, **creating a concentration gradient**.
- **Protons** flow back through **ATP synthase** into the matrix, down the concentration gradient by **Chemiosmosis** (since the **inner/outer membrane are impermeable to protons**), **ATP Synthase** absorbs kinetic energy of the protons, using it to phosphorylate ADP into **ATP** (ADP + Pi → ATP)

3. Oxygen's Role:

- **Oxygen acts as final electron acceptor** to form water. Without it, the **ETC** stops, stopping **proton pumping** and **ATP** production.

Check Diagram below

Oxidative Phosphorylation



VERY IMPORTANT:

Q: Explain how a lack of **oxygen** (anaerobic conditions) affects the oxidative phosphorylation

MS Style answer:

- Since Oxygen is the **final acceptor of electrons**
- The **ETC** stops releasing energy from **excited electrons** for the **proton pump**
- No **protons** pumped into the **intermembrane space**, **no proton gradient** created
- No protons move through **ATP Synthase**, No **Chemiosmosis**, **No ATP Produced**

Q: Describe the main conditions that are required for **pyruvate** to enter the **mitochondrion** by active transport.

Ms Style answer:

- **Mitochondrial membrane** to **be impermeable** to **pyruvate**
- Presence of suitable **transport protein**
- **Oxygen** to be available
- **ATP** for **active transport** against concentration gradient of **pyruvate**

Exam Tip: Oxidative phosphorylation produces **~28 ATP** per glucose—the main ATP source in aerobic respiration, and Substrate linked phosphorylation only produces **4 ATP**:

1. **Glycolysis:** **2 ATP** + (**2 NADH** x 2.5 = **5 ATP**) +

1 NADH ≈ 2.5 ATP

1 FADH ≈ 1.5 ATP

2. **Link Reaction:** **2 NADH** x 2.5 = **5 ATP**) +

3. **Krebs Cycle:** **2 ATP** + (**6 NADH** x 2.5 = **15 ATP**) + (**2 FADH** x 1.5 = **3 ATP**)

Aerobic respiration total ATP = 2 + 5 + 5 + 2 + 15 + 3 = 32 ATP

Q: Explain why the actual net number of ATP molecules produced from aerobic respiration is less than the theoretical number

- **ATP** is used to transport **pyruvate** into the mitochondria by active transport
- Some **protons** **leak out** of the intermembrane space during oxidative phosphorylation Some kinetic energy of protons is **lost as heat**
- **Reduced NAD** may be used for other reactions in the mitochondria

Test Yourself:

Q: Fill this table up:

Stages of Respiration	ATP used	ATP made	Net gain in ATP
Glycolysis			
Link reaction			
Krebs cycle			
Oxidative phosphorylation			
Total			

ALL IMPORTANT ORGANIC MOLECULES IN AEROBIC RESPIRATION

- Glucose is 6C (Glycolysis)
- Triose phosphate and Glycerate phosphate are 3C (Glycolysis)
- Pyruvate is 3C (produced in Glycolysis, goes into Link Reaction)
- Citrate is 6C (Oxaloacetate + Acetyl CoA → Citrate, in Krebs Cycle)
- Acetate/Acetyl CoA is 2C (Link Reaction, goes into Krebs Cycle)
- Oxaloacetate is 4C (gets regenerated in Krebs Cycle)

Q: Outline the roles of named coenzymes in aerobic respiration.

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Q: Explain why the link reaction only occurs when oxygen is available (explain what happens in link reaction and where oxygen is needed)

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Q: Describe the processes that occur in the mitochondrial matrix (with ALL keywords mentioned earlier)

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Now, let's answer the question, Explain Why 1g Fat produce more ATP than 1g Carbohydrates as a respiratory substrate?

- Fats have **more C-H bonds** than carbohydrates, so **more H atoms** than carbohydrates So **more reduced NAD** (NADH) produced
- So **more H atoms released at ETC**, more **protons** to build up **steeper proton gradient in intermembrane space**
- More **chemiosmosis**, more protons passing through ATP synthase

Respiratory Quotient (RQ Value)

Respiratory Quotient: Ratio of volume of carbon dioxide given out divided by volume of Oxygen taken in Per unit time

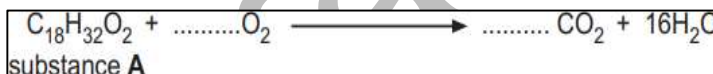
Common RQ values:

- Carbohydrates ≈ 1
- Lipids ≈ 0.7
- Protein ≈ 0.9

$$RQ = \frac{\text{Volume of CO}_2 \text{ out}}{\text{Volume of O}_2 \text{ in}}$$

Test Yourself:

This Fig. Summarises the respiration of an organic substance, A, in aerobic conditions. Complete in the spaces provided.



Calculate the RQ for substance A.

Write your answer to two decimal places. RQ =[1]

Use your calculated RQ value to suggest the group of compounds to which substance A belongs.

..... [1]

***Q: Explain why carbohydrates, lipids and proteins have different relative energy values as substrates in respiration In aerobic conditions. (N18/V2/Q9a)**

- Different substrates have different numbers of hydrogen atoms /C-H bonds Lipids have more C-H bonds in fatty acids than carbs or proteins
- Breaking down of substrates gives H atoms, which reduce NAD/FAD
- Reduced NAD/FAD Release Hydrogen to ETC, which splits into protons and electrons
- More electrons passed to ETC = more energy released to pump protons
- More protons = steeper proton gradient = more chemiosmosis = more ATP produced from lipids

Anaerobic Respiration

Occurs in Anaerobic conditions (absence of oxygen)

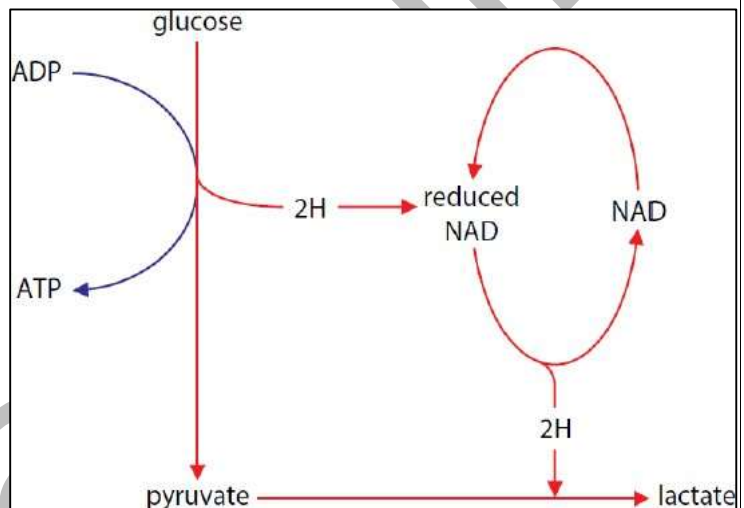
- No **Final Acceptor of electrons and protons**
- **NAD** Can't be regenerated from **reduced NAD**
- **Pyruvate doesn't** enter the mitochondrion, **No link reaction**, no **Krebs cycle**, no **oxidative phosphorylation**

So anaerobic respiration is completely based on Glycolysis in the cytoplasm only

*Anaerobic respiration has 2 pathways

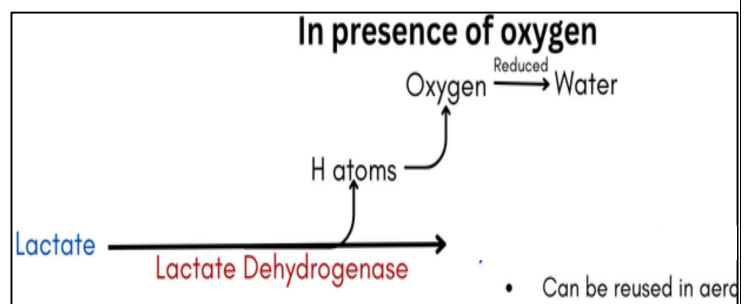
Animal/Bacteria Cells (**Lactate Pathway**):

1. **Pyruvate** accepts **electrons** from **NADH** to form lactate. Building up **oxygen debt**
2. **Regenerates NAD** To be able to continue glycolysis again
3. But creates **oxygen debt** to convert the **lactate** back into **glucose** by reoxidising it
4. Produces **2 ATP** from **glycolysis** (very little amount of energy from **1 glucose**)



When Oxygen is present:

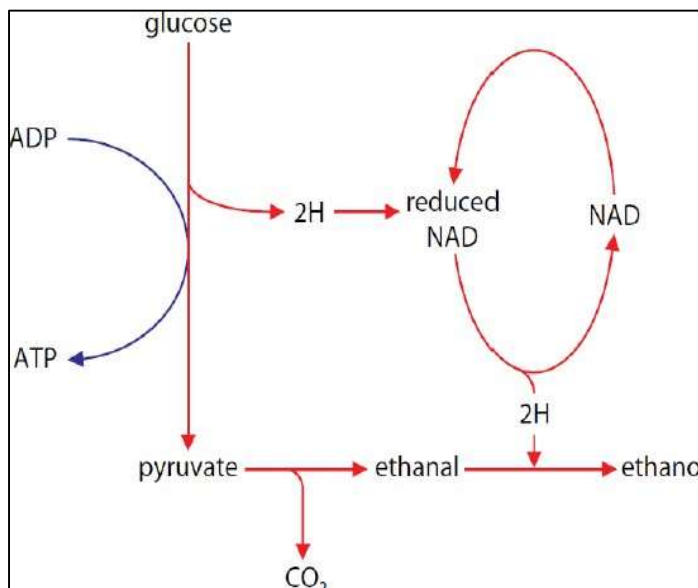
- **Lactate Dehydrogenase** oxidises **Lactate** back into pyruvate
- **Oxygen accepts the H atoms from the lactate and gets reduced into H₂O**
- So, **Lactate pathway** anaerobic respiration is reversible



Yeast/Plant Cells

(Ethanol Pathway / ethanol fermentation):

1. **Glucose** goes through Glycolysis, converted to **Pyruvate**, producing **reduced NAD** and **2 ATP**
2. **Pyruvate** gets **decarboxylated** converts to **ethanal**,
3. **Reduced NAD** releases **H** and **reduces ethanal to ethanol**
4. Releases CO_2 and **regenerates NAD to continue glycolysis**.



Ethanol cannot be converted back to Pyruvate, No oxygen debt

Oxygen debt:

Q: Why is more oxygen required in the body after a session of vigorous exercise?

- To reoxidise **Lactate** in the liver **back to pyruvate**
- To reconvert **pyruvate back into glucose**
- To **reoxygenate haemoglobin**

Differences between Lactate pathway and Ethanol Pathway

Lactate Pathway	Ethanol Pathway
Takes place in Animal and Bacteria cells	Takes place in Yeast and Plant cells
Pyruvate Reduced to Lactate (one step process)	Pyruvate Reduced to Ethanal, THEN Ethanal is reduced to Ethanol (Two step process)
No Decarboxylation	Decarboxylation of Pyruvate first
Reversible Reaction	Irreversible
Lactate Dehydrogenase Can oxidise lactate into Pyruvate	Ethanol Dehydrogenase Can oxidise Ethanol into Ethanal only

Exam Tip: Anaerobic respiration primarily regenerates NAD. ATP comes only from glycolysis by substrate linked phosphorylation.

Test yourself:

Q: Explain why less ATP is produced when yeast respire in anaerobic conditions compared to when yeast respired in aerobic conditions

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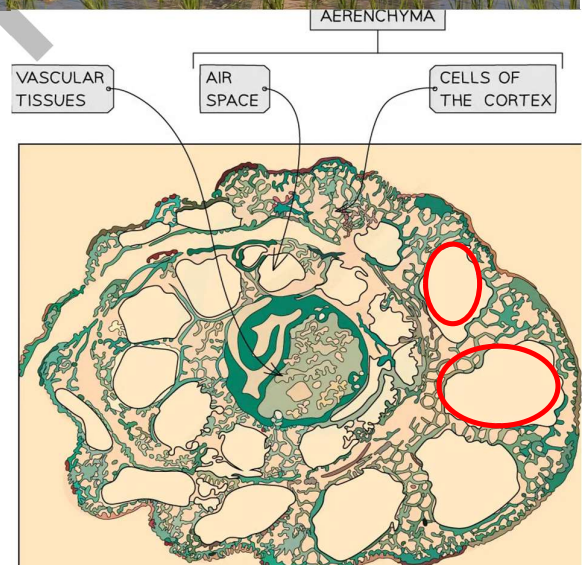
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Adaptations of Rice plants to survive in flooded areas

As the water rises in plants fields, plant roots **don't get enough oxygen** that they need for **aerobic respiration**, plant leaves don't get the **CO₂** they need for **Photosynthesis**.

Rice plants have adapted to surviving in these conditions, as they have:

- **Increased upwards growth rate** so that the leaves remain above water and can **access Oxygen and CO₂**
- **Aerenchyma tissue** in stems and roots
 - A tissue of Air spaces **spreading throughout the plant**
 - **Allows** gases entering the Stomata to travel to the stem and roots **that are submerged underwater, allowing continuing of** aerobic respiration
- When **Aerobic isn't enough**, rice plant tissues can undergo **ethanol fermentation**
 - As they tolerate the **high levels of toxic ethanol**
 - Due to High concentrations of **Ethanol Dehydrogenase**, which breaks down **ethanol**



End Of Chapter

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